

Name: Sudha Sree, Yerramsetty

Title of paper: “Counterfactual Fairness” by Matt Kusner, Joshua Loftus et al.

Summary:

This paper introduces a new causal framework for achieving fairness in machine learning called “Counterfactual Fairness”. The authors argue that traditional statistical definitions of fairness, such as demographic parity, equality of opportunity or odds, or fairness through unawareness fail to capture the underlying causal relationships that generate bias in data. Their core idea is that a prediction is fair if it would hold true for an individual in both the real world and a counterfactual world in which that person's protected attribute (e.g., race, gender) has been changed. This approach relies on Judea Pearl’s causal modeling framework to explicitly model how protected attributes affect other variables and outcomes. Using causal diagrams and counterfactual reasoning, they formalize fairness as the prediction's invariance under such counterfactual modifications.

The paper mathematically defines this notion, provides a method to implement it, and demonstrates it using a real-world case study predicting law school performance. They prove that, while implementing counterfactual fairness somewhat reduces prediction accuracy, it prevents discrimination due to historical bias. The experiments highlight how causal reasoning enables fairer predictions compared to traditional methods that simply ignore sensitive attributes. Overall, the paper presents a compelling case that genuine fairness requires causal reasoning rather than mere statistical independence, and it gives a formal, testable definition to guide future algorithmic fairness research.

Two points that you like about the paper (strengths):

- The paper’s greatest strength is that it brings causal inference into fairness research, addressing the limitations of purely statistical approaches. By reasoning about how protected attributes cause other variables, the authors provide an interpretable definition of fairness that aligns with human moral reasoning about “what would have happened if things were different”.
- The research paper not only discusses a theoretical notion, but it also includes an algorithmic approach and a real-world example (law school success prediction). This makes the idea more tangible by demonstrating that counterfactual fairness can be used in practice and empirically verified using causal models and probabilistic programming.

Two points that you did not like about the paper (weaknesses):

- The framework makes the assumption that all variables, including unobserved latent factors, can be correctly specified in terms of causal relationships. However, in practice, constructing an accurate causal model is difficult and often subjective, limiting the practical applicability of counterfactual fairness in complex real-life situations.
- The proposed approach in this paper uses MCMC sampling and structural modeling, both of which can be computationally expensive and theoretically difficult for users to

understand. The framework's mathematical difficulty might discourage its use outside of research contexts, particularly when simpler fairness metrics can be applied with more ease.

Detailed comments and questions:

- The paper convincingly argues that fairness should be defined at the individual level through counterfactual reasoning rather than group-level parity. Can such an individual-level fairness scale be used effectively in large-scale decision-making systems (e.g., credit scoring)?
- The trade-off between fairness and accuracy is acknowledged but not deeply explored. Future research might further better quantify this trade-off, however under what conditions can we attain high fairness and predictive power?
- Overall, the paper is a foundational contribution that integrates fairness and causality; yet it leaves open practical challenges such as model identifiability, scalability, and real-world implementation, which future research will have to address.